BDMM 2026: HW Answers

Submit your answers by April 3 if you would like the grade before the exam.

Final deadline is April 10, 2026.

Please do not copy the task description into this form, only your answers.

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HW Group: 23

Your role (chose one): Part D

List the overall transformations to the database **you group** made to the database: e.g. embedding / referencing choices, pattern uses, indexes, any further redesign and restructuring. [*only one person per group needs to answer this for the whole group*]

[max 450 words]

We redesigned the original Airbnb dataset from a monolithic listing-oriented structure into a cleaner MongoDB model with a clearer separation between core listing data, stable entities, long text, and high-growth transactional content. The goal was to keep the main listings collection compact and query-friendly while moving bulky or unbounded information into separate collections. In particular, host information was extracted into a dedicated hosts collection, rich textual fields such as description and house rules were moved to listing_details, and embedded reviews were externalized into a separate reviews collection. This reduced document bloat, removed repeated host fields across listings, and avoided continuously growing review arrays in the main listing documents.

During migration, missing placeholders and invalid values were normalized to null, numeric and date fields were cast to consistent types, geographic coordinates were validated, suspicious values were checked, and duplicate reviews were removed using a composite deduplication rule. The final validation confirmed that all 5,555 listings were preserved, 5,497 listing_details documents were created, 5,104 unique hosts were extracted, and 149,791 reviews were stored separately after removing one duplicate review record.

The remodeled design applies standard document-database modeling principles. A small host snapshot and review statistics remain embedded in listings for fast access, *while full host records, long textual details, and reviews were separated into referenced collections. Reviews were modeled through referencing because they form an unbounded one-to-many relationship, and listings follow a subset-style logic by keeping only the most useful operational and analytical fields in the main collection. This keeps frequently accessed listing documents smaller and more efficient to read, while preserving the full review history in a dedicated collection. These choices are consistent with the document-model guidance of embedding small, frequently co-read data and linking high-volume relationships on the many side.*

To support query performance, indexes were created on important access paths such as market, property type, room type, price-related fields, host identifiers, review-related fields, geospatial location, review date combinations, and text-search fields. JSON schema validators were also applied to listings, reviews, hosts, and listing_details to enforce the remodeled structure. Overall, the database was transformed from a raw import into a more scalable, consistent, and analytics-ready document model.

Question 1. Assumptions *[max 350 words]*

Professional hosts were defined as hosts managing more than one listing in the remodeled database; amateur hosts were hosts managing exactly one listing. The analysis was restricted to Hong Kong, Montreal, and Barcelona, as requested. Effective price was calculated for a stay of two guests over seven nights, using the listing price plus applicable fees. The notebook used the following logic:
effective_price_pppn = (7 × price + cleaning_fee + 7 × extra_people × max(0, 2 − guests_included)) / 14.

Thus, the metric represents the effective price per person per night after spreading one-off cleaning fees across the stay and adding extra-person charges only when the assumed two guests exceeded guests_included. Listings with missing or invalid pricing inputs were excluded automatically by the cleaning pipeline, while the remodeled database preserved valid zero-missing logic through typed numeric fields.

For the visualization, hosts were compared within each city using box plots. To keep the boxes interpretable despite extreme luxury outliers, the chart was visually capped at the 99th percentile. However, the true minimum and maximum values were still retained in the summary statistics and used in the interpretation. This keeps the figure readable without discarding the real range of observed prices.

Question 1. Visual

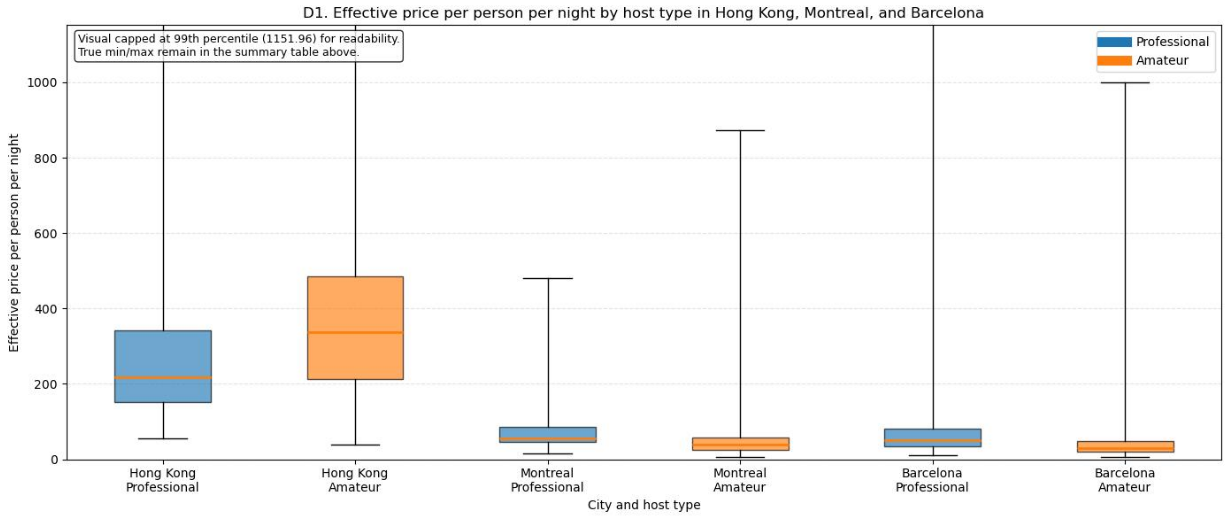

Figure 1. Effective price per person per night by host type in Hong Kong, Montreal, and Barcelona.

Question 1. Interpretation *[max 350 words]*

Across the pooled three-city sample, professional hosts charge more on the median in Barcelona and Montreal.

In Hong Kong, amateur hosts are substantially more expensive on the median. The median effective price is 338.21 for amateurs versus 216.79 for professionals, a difference of 121.43 in favor of amateurs. Dispersion is also wider for amateurs: the interquartile range is 273.71 compared with 191.54 for professionals. This suggests that Hong Kong amateur listings include a broader and more expensive upper segment.

In Montreal, the pattern reverses. Professional hosts have a median of 54.50, compared with 38.93 for amateurs, so professionals are higher by 15.57. The professional IQR is also wider, 42.21 versus 32.63, indicating more heterogeneity among professional operators. Barcelona shows a similar result: professionals have a median of 51.11 and amateurs 30.00, a gap of 21.11, with a wider IQR for professionals as well, 46.82 versus 28.96.

Overall, being a professional host is associated with higher prices in Montreal and Barcelona, but lower prices in Hong Kong. Therefore, professionalism does not have a uniform effect across markets. The box plots also show that price distributions are highly skewed, so the median and IQR are more informative than the mean for comparing host segments.

Question 2. Assumptions *[max 350 words]*

Customer satisfaction was measured using review_scores_rating. When a host managed multiple listings, the notebook first aggregated ratings to the host level by computing a weighted average rating across that host's listings, using review count as the weight. This reduces the influence of hosts with very few reviewed listings.

The original host attributes were then converted into binary analytical features. Superhost status was taken directly from host_is_superhost. Response time was simplified into fast responder versus not fast responder. Host verification was taken from host_is_verified. Response rate was converted into high response rate versus not high response rate. Cancellation policy was collapsed into relaxed cancellation versus stricter cancellation. These transformations were necessary because the assignment requires a binary comparison for the double-lollipop chart.

From a database-design perspective, the raw listing-to-host aggregation was then materialized into a host_satisfaction_summary collection using $merge, so repeated analytical reads would not need to recompute the same host-level metrics from listings. The performance comparison therefore contrasts the raw aggregation logic with the optimized computed-summary design.

Question 2. Visual of data

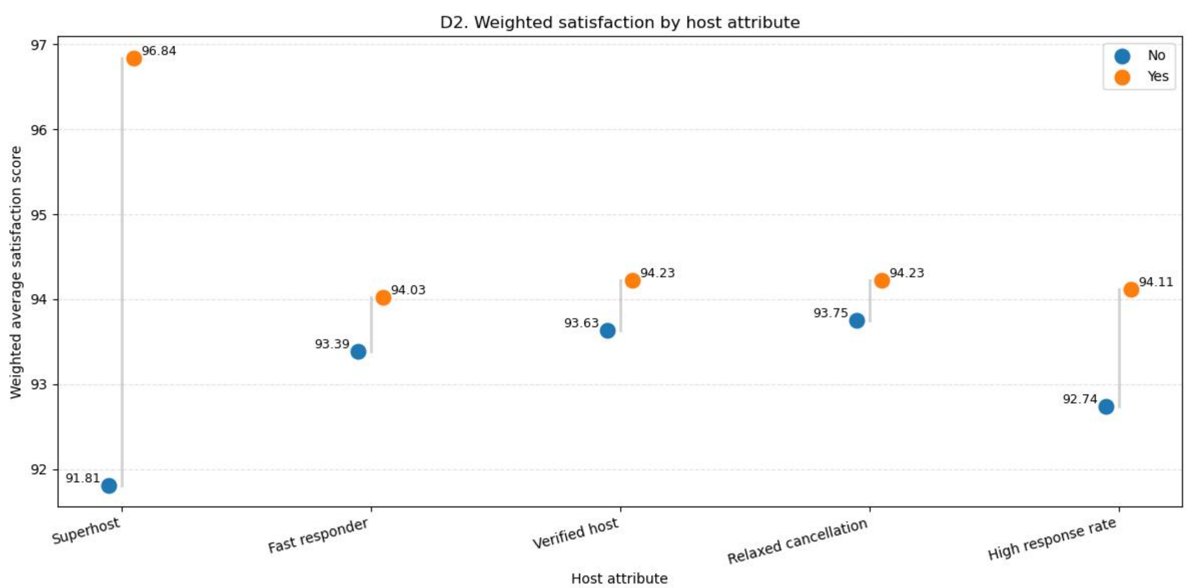

Figure 2. Weighted host-level satisfaction score by binary host attribute.

Question 2. Query Performance Comparison

Version	Avg ms	Docs examined	Keys examined	Main stages	Memory
Raw pipeline (before extra index)	1284.94	5,555	5,555	PROJECTION_DEFAULT -> FETCH -> IXSCAN -> $addFields -> ...	668,272 B
Raw pipeline (after supporting index)	1444.81	5,555	5,555	PROJECTION_DEFAULT -> FETCH -> IXSCAN -> $addFields -> ...	668,272 B
Computed host summary pipeline	65.84	3,797	3,797	PROJECTION_SIMPLE -> FETCH -> IXSCAN -> $facet -> ...	352 B

Note: Execution time was benchmarked over five repeated aggregate runs. Documents examined, keys examined, stage structure, disk usage, and maximum accumulator memory were obtained from explain output. The additional supporting index did not improve this workload, as both raw versions still examined 5,555 documents and 5,555 keys, with the same maximum accumulator memory of 668,272 bytes and no disk spill. In contrast, the computed host_satisfaction_summary pipeline reduced the workload to 3,797 documents and keys examined, only 352 bytes of accumulator memory, and no disk spill. Minor execution-time variation across runs is expected under normal conditions.

Question 2. Interpretation *[max 450 words]*

The weighted host-level results indicate that Superhost status is the strongest positive differentiator of satisfaction. Hosts without Superhost status have a weighted average score of 91.81, while Superhosts reach 96.84, a gap of 5.03 points. The second-largest gap appears for high response rate, 94.11 versus 92.74, a difference of 1.37 points. Fast response time, verification, and relaxed cancellation are also associated with higher satisfaction, but their effects are smaller, at roughly 0.47 to 0.64 points.

The performance comparison reinforces the argument for the computed pattern. In the current notebook output, the raw listing-level pipeline averaged 1284.94 ms before the extra supporting index and 1444.81 ms after it. In both cases, MongoDB examined the same 5,555 documents and 5,555 keys, while maximum accumulator memory remained unchanged at 668,272 bytes, indicating that the extra index did not reduce the main aggregation cost. By contrast, querying the materialized host_satisfaction_summary collection reduced average runtime to 65.84 ms, with 3,797 documents and 3,797 keys examined and only 352 bytes of accumulator memory. This makes the computed version about 19.5 times faster than the original raw pipeline. Therefore, the main optimization came from changing the data-access pattern through a computed summary, not from adding another supporting index to the raw listing collection.

Question 3. Assumptions *[max 350 words]*

Popularity was defined as future unavailability, under the assumption that for professional hosts unavailable days are more likely to represent real bookings than ad hoc blocking. The notebook restricted the analysis to professional hosts only and created a persistent popularity_score attribute in the listings collection.

The score was computed as a weighted combination of future unavailability across the next 30, 60, 90, and 365 days:
popularity_score = 100 × [0.40 × (1 − availability_30/30) + 0.30 × (1 − availability_60/60) + 0.20 × (1 − availability_90/90) + 0.10 × (1 − availability_365/365)].

This weighting emphasizes short-term demand more heavily than distant-horizon availability. The score was then stored in the database and indexed for repeated analytical access.

For satisfaction, fixed and interpretable bands were used instead of sample-dependent tertiles: Low < 80, Medium = 80 to 95, and High > 95 on the 0–100 review score scale. The comparison was visualized for the three most common property types among professional listings: Apartment, Condominium, and Serviced apartment.

Question 3. Visual

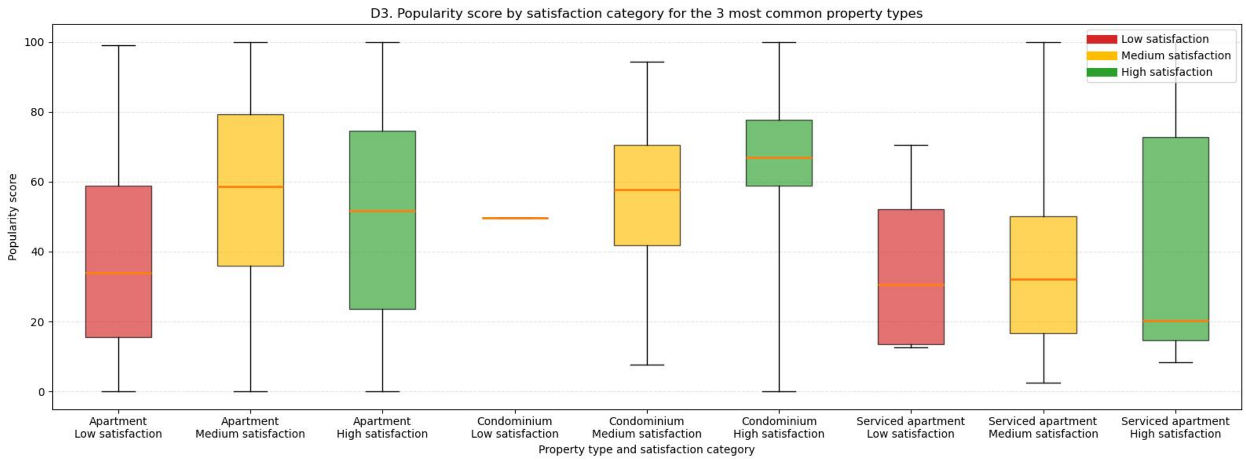

Figure 3. Popularity score by satisfaction category for the three most common professional-host property types.

Question 3. Interpretation *[max 450 words]*

At the aggregate level, popularity increases with satisfaction. Among professional-host listings, the median popularity score is 33.84 for low satisfaction, 57.64 for medium satisfaction, and 58.78 for high satisfaction. This supports a positive overall relationship between guest satisfaction and future unavailability, hence between satisfaction and likely demand.

The pattern is clearest for Condominium, where the median popularity rises from 49.69 in the low-satisfaction group to 57.64 in the medium group and 66.95 in the high group. Apartment also shows a broadly positive pattern: low satisfaction has the lowest median popularity at 33.84, while medium satisfaction is highest at 58.53 and high satisfaction remains elevated at 51.68. This suggests that better-reviewed apartments tend to be less available, although the medium and high groups still overlap substantially.

Serviced apartment is less stable: the median popularity is 30.61 for low satisfaction, 32.18 for medium satisfaction, and only 20.16 for high satisfaction. However, this category has small group sizes, especially in the high-satisfaction segment, so the result should be interpreted cautiously. Overall, the box plots indicate that satisfaction is positively associated with popularity for the larger property types, but the relationship is not deterministic. Location, pricing, and host strategy likely remain important additional drivers of demand.